

El-Shorouk Academy
The Higher Institute of Engineering
Communications and Computers Engineering Department

Project Title:

**Speed and Direction Control for Dual DC Motor using
Timer 555 and H-Bridge**

Prepared by

No.	Name	Sec
1	Ali El-Sayed	6
2	Ali Mohamed Abdelsalam	6
3	Omar Mohamed	6
4	Philopateer Charl	6
5	Nour Eldeen Nashat	10

Supervised by
Dr. Nabil Abd-Rabou

Abstract

The speed and Direction Control for Dual DC Motor using Timer 555 and H-Bridge is a fundamental electronic circuit that finds applications in Simple differential drive robots, educational demonstrators for PWM and motor control principles, systems requiring simple motion control. This report provides a comprehensive explanation of the circuit design, components, working principles, and applications. By understanding the intricacies of the H_Bridge & PWM, electronics enthusiasts and professionals can effectively use it in diverse applications.

Acknowledgement

To

Dr. Nabil Abd Rabou

At the beginning of our report, we would like to show that we are grateful for your great efforts. We appreciate all of it, from explaining the theoretical meaning of all electronic components we have used to the practical part, until we have reached this result right now.

Contents

Abstract.....	2
Acknowledgement	2
Introduction.....	6
Circuit diagram	6
Circuit Design and Operation	7
Objective	7
How Speed Control of DC Motor Circuit Works?	7
Components Guide.....	8
1- DC Geared Motor Dual Shaft 3:12V DC	8
2- NE555 Timer IC	8
3- Tip142 NPN Transistor	8
4- Tip147 PNP Transistor	8
5- BC337 NPN Amplifier Transistor	8
6- Dual Metal Potentiometer 1M Ω	9
7- Toggle Switch 3 Pins - 2 Position	9
8- Mixed LED 5mm	9
-9 1N4007 Diode.....	9
-10 Capacitors.....	9
11- Resistors.....	9
Working Principle.....	10
Speed Control:	10
Direction Control:	10
Combined Operation:	10
Key Equations	10
1. PWM Frequency Calculation:	10
Simulation and Testing.....	11
The First Step	11
Schematic Capture	11
.....	11
Output from the Circuit	12
The Second Step	13
Test using bread board	13
Testing PWM Part Of the Circuit	13
Testing The Whole Circuit	13
PCB Design & Implementation.....	14
The Third Step	14
PCB Schematic	14

The Fourth Step	15
Designing the PCB Layout	15
The Fifth Step	16
3D visualization	16
The Sixth Step.....	16
Exporting CAM Files.....	16
The Seventh Step.....	17
Ordering Our PCB.....	17
Getting Our PCB Done.....	17
The Eighth Step	18
Components Soldering.....	18
.....	18
The ninth Step	18
Finishing The project.....	18
Advantages and Limitations	19
Advantages.....	19
Limitations.....	19
Applications	19
Enhancements.....	19
Conclusion.....	19

Table of Figures

Figure 1 Circuit diagram-----	6
Figure 2: DC Geared Motor Dual Shaft -----	8
Figure 3: NE555 Timer IC-----	8
Figure 4: Tip142 NPN Transistor -----	8
Figure 5: Tip147 PNP Transistor-----	8
Figure 6: BC337 NPN Amplifier Transistor -----	8
Figure 7: Dual Metal Potentiometer 1M Ω -----	9
Figure 8: Toggle Switch 3 Pins -----	9
Figure 9: LED 5mm-----	9
Figure 10: 1N4007 Diode-----	9
Figure 11: Capacitors-----	9
Figure 12: Resistors-----	9
Figure 13 proteus Simulation Schematic-----	11
Figure 14 PWM Testing board -----	13
Figure 15 PWM Duty Cycle Wave Form -----	13
Figure 16 Speed & Direction Circuit Testing -----	13
Figure 17 PCB Schematic Design -----	14
Figure 18 PCB Layout -----	15
Figure 19 PCB 3D visualization -----	16
Figure 20 Exportation of Gerber Files-----	16
Figure 21 PCB Order-----	17
Figure 22 PCB Design VS Reality -----	17
Figure 23 PCB After Soldering-----	18
Figure 24 Simple Car Robot Application-----	18

Introduction

Direct Current (DC) motors are essential actuators in robotics, automation, and electronic systems due to their simplicity and reliability. Precise control over motor speed and direction is crucial in applications such as wheeled robots, conveyor systems, and remote-controlled vehicles. This project presents a cost-effective and practical method to control two independent DC motors using a 555 Timer IC for speed control via Pulse Width Modulation (PWM) and an H-Bridge circuit for direction control. The system offers an efficient solution for various mechatronics and automation projects without the need for complex microcontrollers. This report explores the theory, design, and implementation of this dual motor control system.

We invite you to join us on a journey through the different phases of our project. This report outlines each step we took to bring the system to life—from simulation and testing to designing the PCB layout and completing the final circuit connections.

Circuit diagram

Speed and Direction Control for Dual DC Motor using PWM and H-Bridge

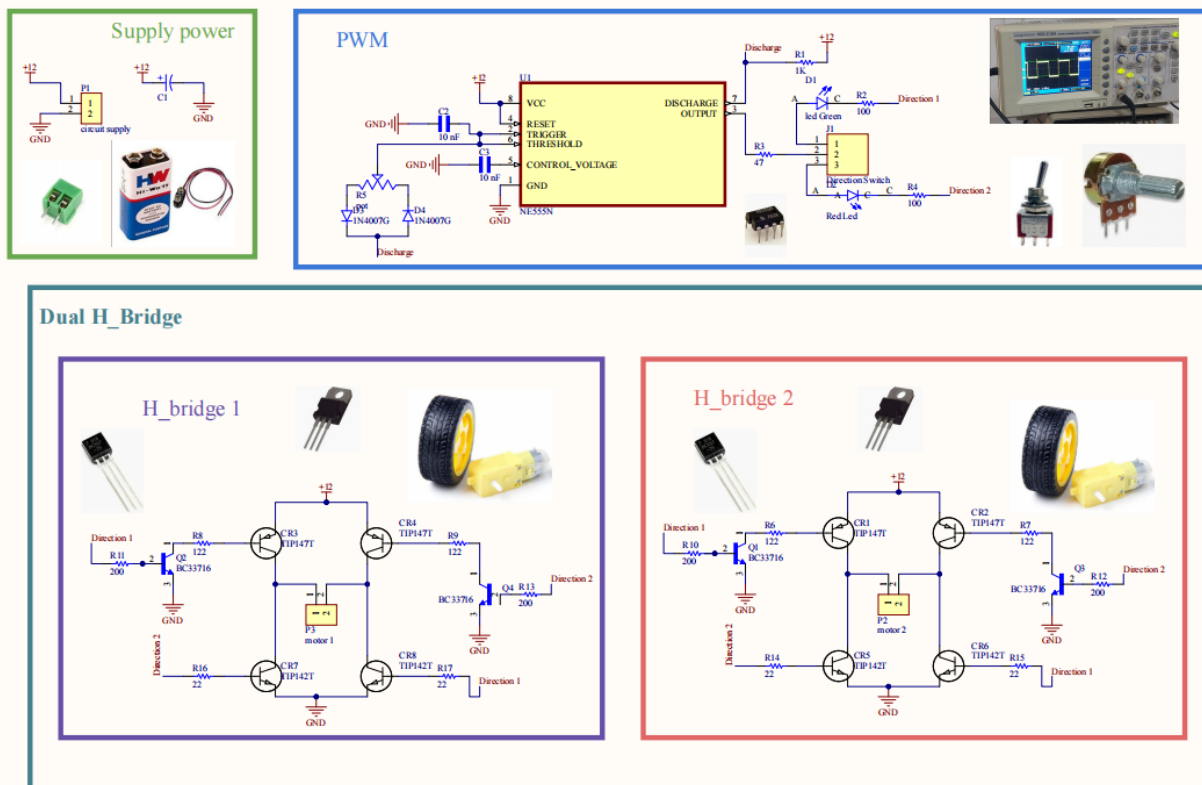


Figure 1 Circuit diagram

Circuit Design and Operation

The system design is split into two main control functionalities: speed control and direction control.

1. **Speed Control using 555 Timer:** The NE555 Timer IC is configured in astable mode, producing a continuous square wave. By adjusting the resistors and a capacitor in the timing network, the frequency and duty cycle of the output can be manipulated. The duty cycle, which determines how long the output stays high during each cycle, directly affects the power delivered to the motors. A higher duty cycle means the motor receives more voltage on average, hence spinning faster. A potentiometer is included to manually vary the duty cycle and thus adjust motor speed.
2. **Direction Control using H-Bridge:** The H-Bridge is made up of four power transistors (two NPN and two PNP for each motor) arranged in an H configuration. The motors are placed in the center of this “H.” By toggling which diagonal pair of transistors is activated, the current direction through the motor is reversed, changing the direction of rotation. SPDT switches are used to control which transistor pairs are active, allowing manual reversal of motor direction.

Objective

- To control the **speed** of two DC motors using **PWM (Pulse Width Modulation)** generated by the **555 timer**.
- To control the **direction** of rotation of the motors using an **H-Bridge circuit**.
- To **simulate** and test the circuit before implementation.
- To design a **PCB layout** for the final build.

How Speed Control of DC Motor Circuit Works?

In this circuit, the DC motor is operated by a 555 integrated circuit. The IC 555 in this circuit is being operated in astable mode, which produces continuous HIGH and LOW pulses.

In this mode, the 555 IC can be used as a pulse width modulator with a few small adjustments to the circuit. The frequency of operation of the circuit is provided by the passive parameters of resistances and capacitors attached to it.

Components Guide

1- DC Geared Motor Dual Shaft 3:12V DC

The DC motor being driven. Its speed is controlled by the **PWM** from the 555 timer, and its direction is controlled by the transistor arrangement in the **H-Bridge**.



Figure 2: DC Geared Motor Dual Shaft

2- NE555 Timer IC

This is a 555 Timer IC configured in **Astable mode**. Its primary function here is to generate a continuous square wave signal. The duty cycle (the percentage of time the signal is HIGH in one cycle) of this square wave can be varied, creating a Pulse Width Modulation (PWM) signal. This PWM signal is used to control the motors' speed.



Figure 3: NE555 Timer IC

3- Tip142 NPN Transistor

Function: Acts as a low-side switch (connects motor to **ground**).

In H-Bridge: Two are on the bottom side of the bridge, controlling current from motor to ground.



Figure 4: Tip142 NPN Transistor

4- Tip147 PNP Transistor

Function: Acts as a high-side switch (connects motor to **+V**).

In H-Bridge: Two are on the top side of the bridge, controlling current from power supply to the motor.

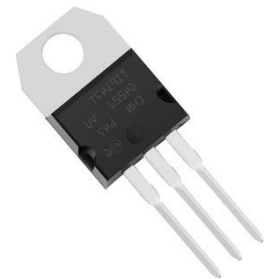


Figure 5: Tip147 PNP Transistor

5- BC337 NPN Amplifier Transistor

Function: These small signal transistors act as drivers for the TIP147s.

They allow the low current **PWM** or logic signal to switch the high-power TIP147 transistors **ON/OFF**.



Figure 6: BC337 NPN Amplifier Transistor

6- Dual Metal Potentiometer 1M Ω

This variable resistor, along with R1, D1, D2, and C2, allows you to adjust the timing of the 555 timer's charge and discharge cycles. Specifically, turning the knob changes the resistance, which alters the duty cycle of the PWM signal coming out of the 555 (Pin 3). A higher duty cycle corresponds to higher average voltage to the motors.



Figure 7: Dual Metal Potentiometer 1M Ω

7- Toggle Switch 3 Pins - 2 Position

A Single-Pole Double-Throw switch. This switch is used to select the direction of rotation for both motors simultaneously. Flipping the switch changes the logic state sent to the driver transistors.



Figure 8: Toggle Switch 3 Pins

8- Mixed LED 5mm

The LEDs D3 and D4 provide a visual confirmation of which direction signal (controlled by SW1 via Q4 and Q5) is currently active, corresponding to the direction the motors are set to turn. One LED lights up for "Forward" (or Direction A), and the other lights up for "Reverse" (or Direction B).



Figure 9: LED 5mm

9- 1N4007 Diode

They protect the circuit from **reverse voltage spikes**, especially caused by **inductive loads** like motors when they stop suddenly.



Figure 10: 1N4007 Diode

10- Capacitors

Value: 3300 μ F/16v - 10nF/50v

They define the **frequency and duty cycle of the PWM** signal used to control motor speed.

Work together with resistors to shape the timing behavior.



Figure 11: Capacitors

11- Resistors

Value: 22 Ω - 47 Ω - 100 Ω - 120 Ω - 200 Ω

Ensure the transistors switch properly without drawing too much current from the control signal.



Figure 12: Resistors

Working Principle

Speed Control:

- Each motor is connected to a separate 555 timer.
- By adjusting the potentiometer, the duty cycle of the PWM signal changes.
- A higher duty cycle provides more power to the motor, increasing its speed.

Direction Control:

- The H-Bridge driver receives logic signals to control the current flow direction.
- Depending on the input combination, the motor rotates forward or reverse.

Combined Operation:

- The PWM output from the 555 timer is fed into the Enable pin of the H-Bridge.
- The H-Bridge input pins (IN1/IN2, IN3/IN4) are controlled by logic buttons or switches.

Key Equations

1. PWM Frequency Calculation:

The frequency of the PWM signal (f_{PWM}) generated by the 555 timer in astable mode is given by:

$$1/\ln(2) = 1.44$$

$$t_{\text{h}} = 0.693 * R1 * C$$

$$t_{\text{l}} = 0.693 * R2 * C$$

$$T = t_{\text{h}} + t_{\text{l}} = 0.693 * (R1 + R2) * C$$

$$f_{\text{PWM}} = 1/T = 1.44 / ((R1 + R2) * C)$$

2. Duty Cycle (D) Calculation:

$$D = t_{\text{H}}/T$$

$$D = ((R1 + R_{\text{pot}}) / (R1 + R2 + R_{\text{pot}})) * 100$$

3. Motor Average Voltage (V_{avg}):

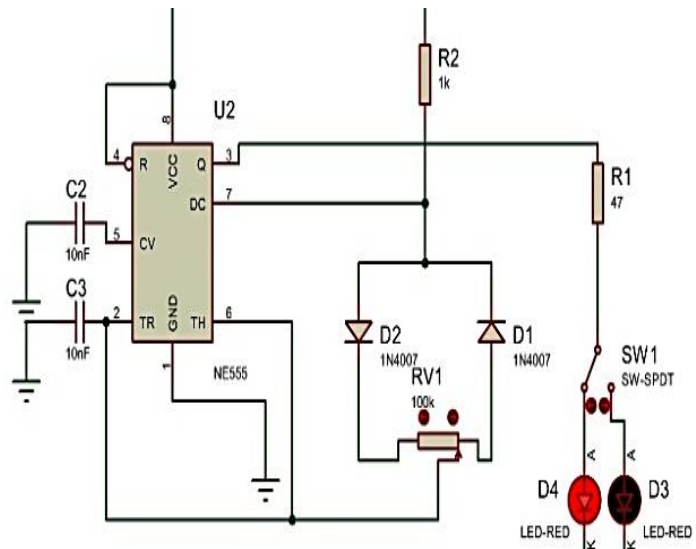
$$V_{\text{avg}} = D * V_{\text{supply}}$$

4. Effective Motor Voltage:

$$V_{\text{motor}} \approx V_{\text{cc}} * (D / 100)$$

5. Motor Power:

$$P = V * I$$



6. H-Bridge Direction Logic (Basic):

- Forward: IN1 = HIGH, IN2 = LOW
- Reverse: IN1 = LOW, IN2 = HIGH
- Brake: IN1 = IN2 = HIGH or LOW

Simulation and Testing

The design was created and tested using Proteus simulation software. The 555 timer was verified for stable PWM output using virtual oscilloscopes. Motor operation was tested by observing motor direction changes and speed variations based on the PWM duty cycle. Different resistor and capacitor values were tested to optimize the response and control range.

The First Step

Schematic Capture

We had to do the simulation for our project first to make sure that the circuit is clearly working and to understand more about circuit components.

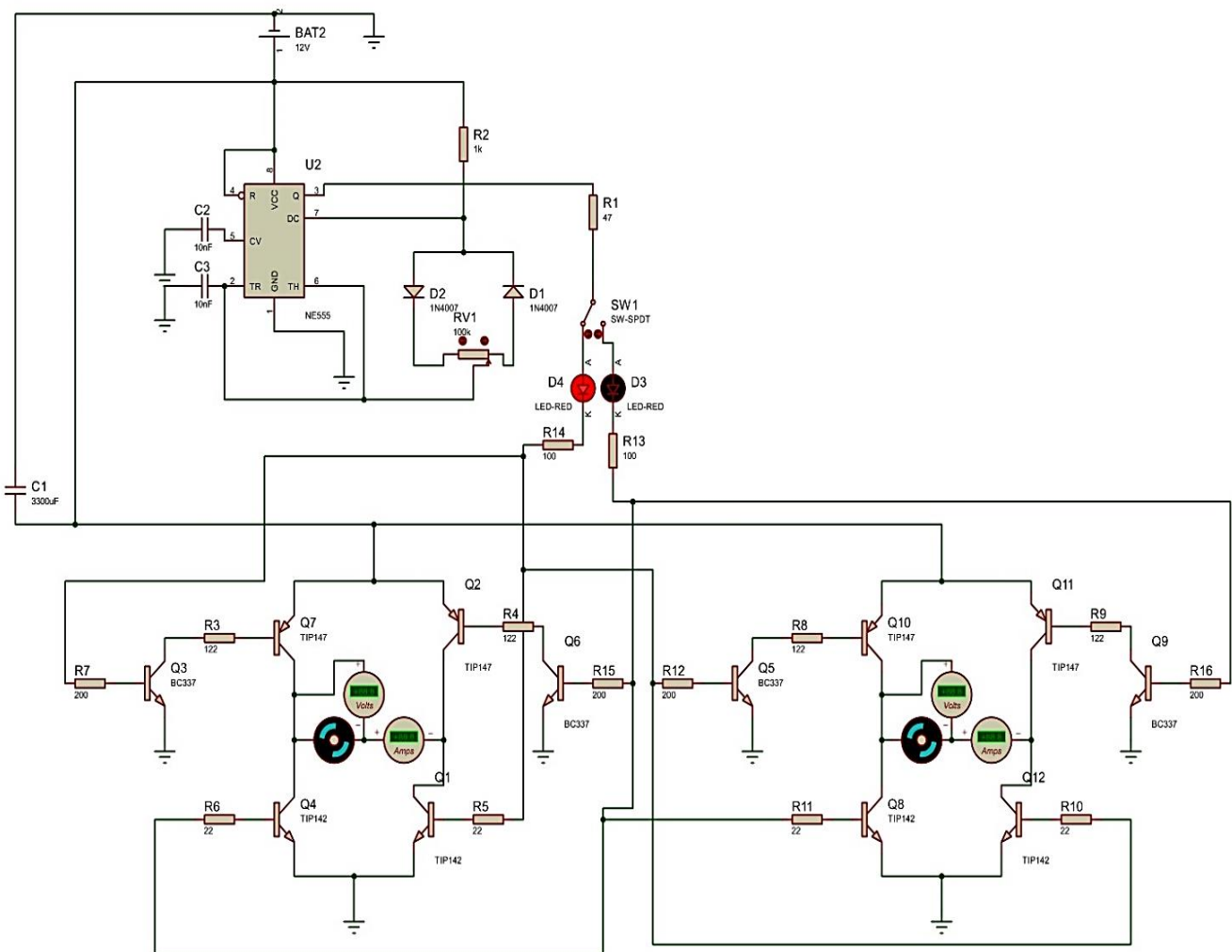


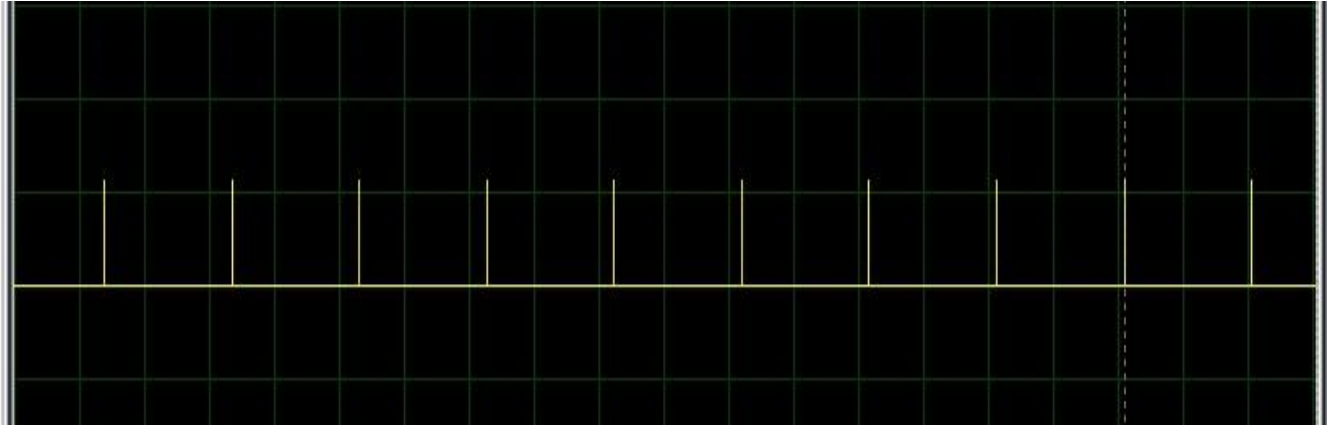
Figure 13 proteus Simulation Schematic

Output from the Circuit

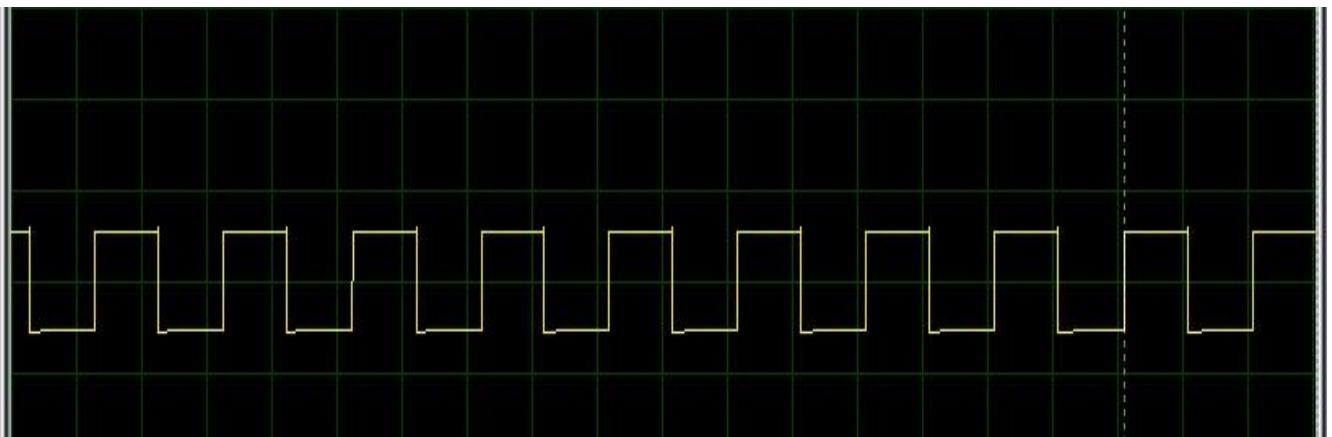
The circuit generated a stable PWM signal with an adjustable duty cycle, controlling the motor's speed accurately. The motor responded well to changes in the potentiometer's position, demonstrating the circuit's capability to provide real-time speed control.

PWM signal observed at Pin 3 of 555 Timer was stable at a duty cycle ranging between 20%-80% based on potentiometer adjustment.

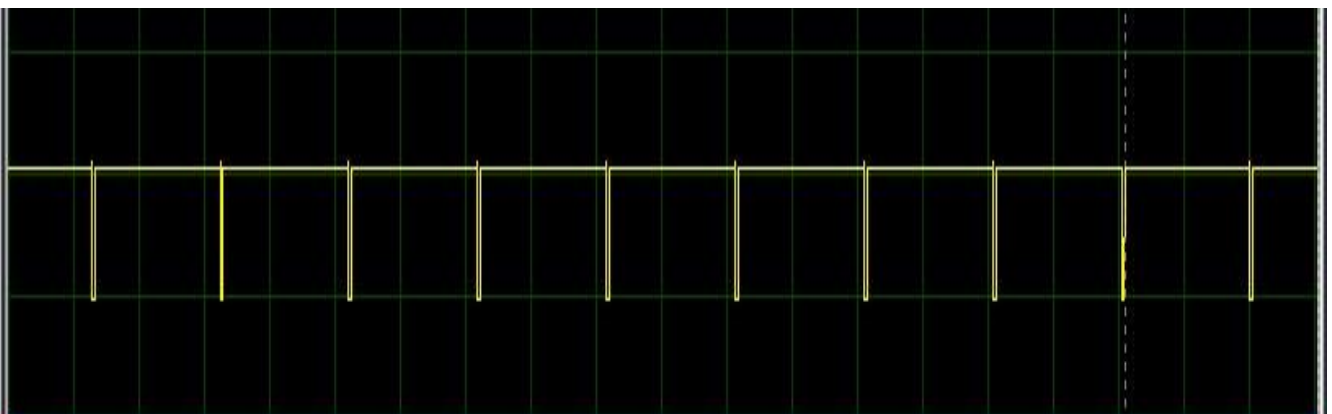
➤ **0% k Ω :**



➤ **50% k Ω :**



➤ **100% k Ω :**



The Second Step

Test using bread board

After the simulation, the circuit was assembled on a breadboard to evaluate real-world performance. The 555 Timer section was connected to a potentiometer to adjust the duty cycle and observe speed changes in the motors. Each H-Bridge circuit was tested using SPDT switches to toggle motor direction. The setup helped identify issues like loose connections and component heating, which were resolved before moving to PCB design.

Testing PWM Part Of the Circuit

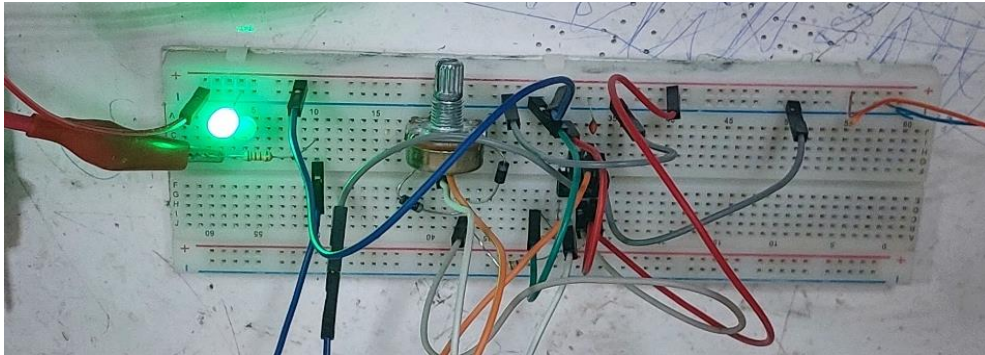


Figure 14 PWM Testing board

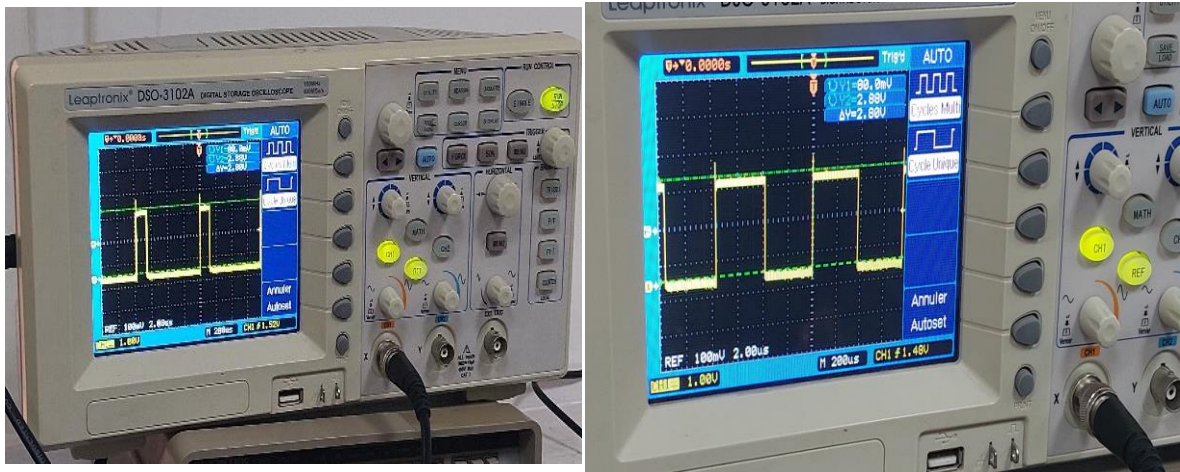


Figure 15 PWM Duty Cycle Wave Form

Testing The Whole Circuit

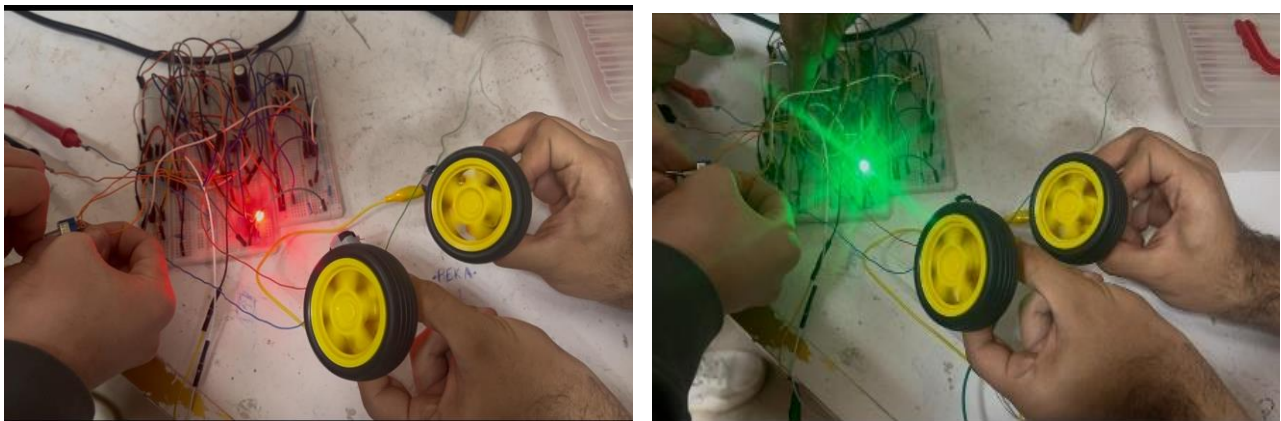


Figure 16 Speed & Direction Circuit Testing

PCB Design & Implementation

The Third Step

PCB Schematic

Speed and Direction Control for Dual DC Motor using PWM and H-Bridge

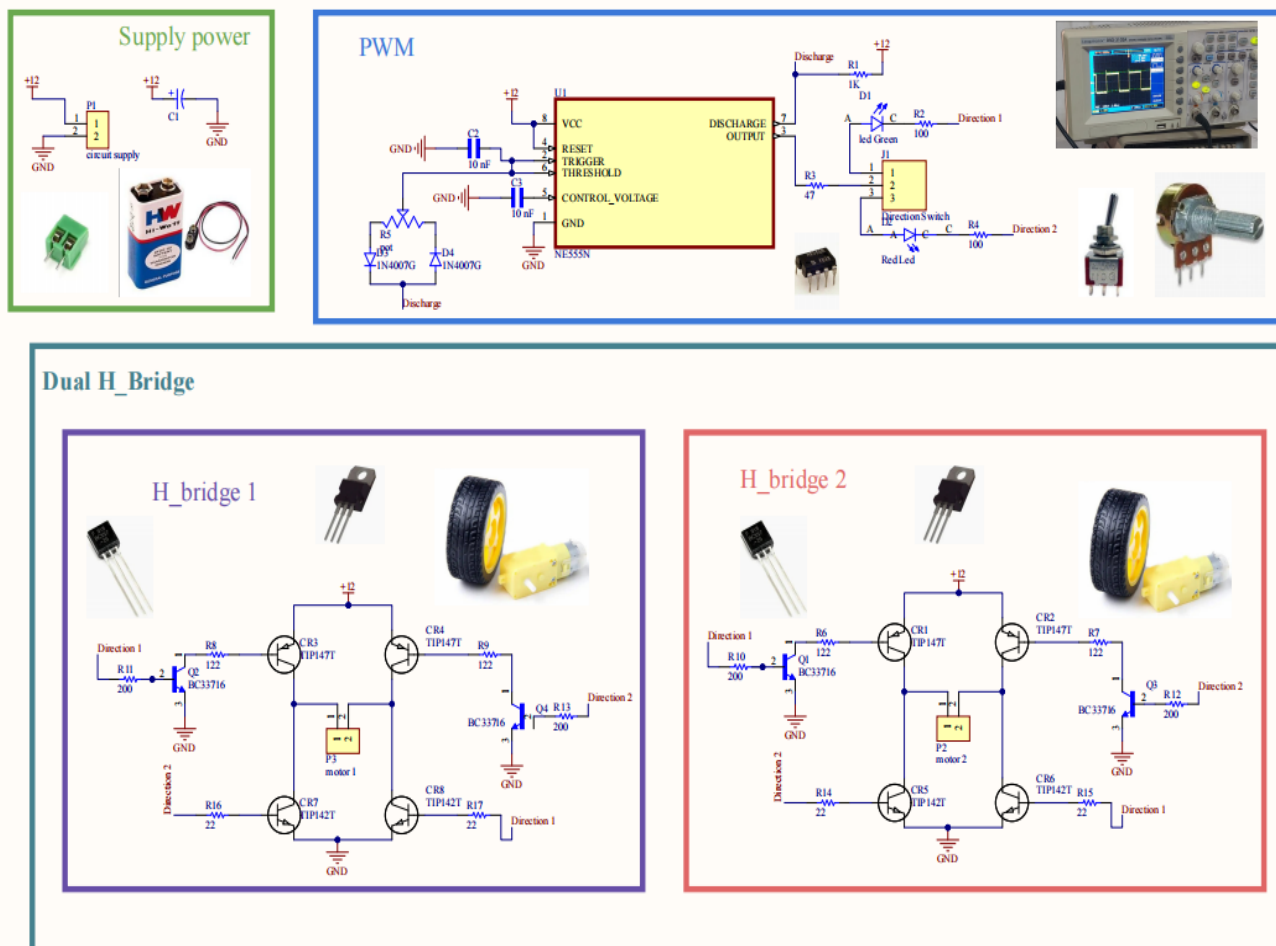


Figure 17 PCB Schematic Design

The Fourth Step

Designing the PCB Layout

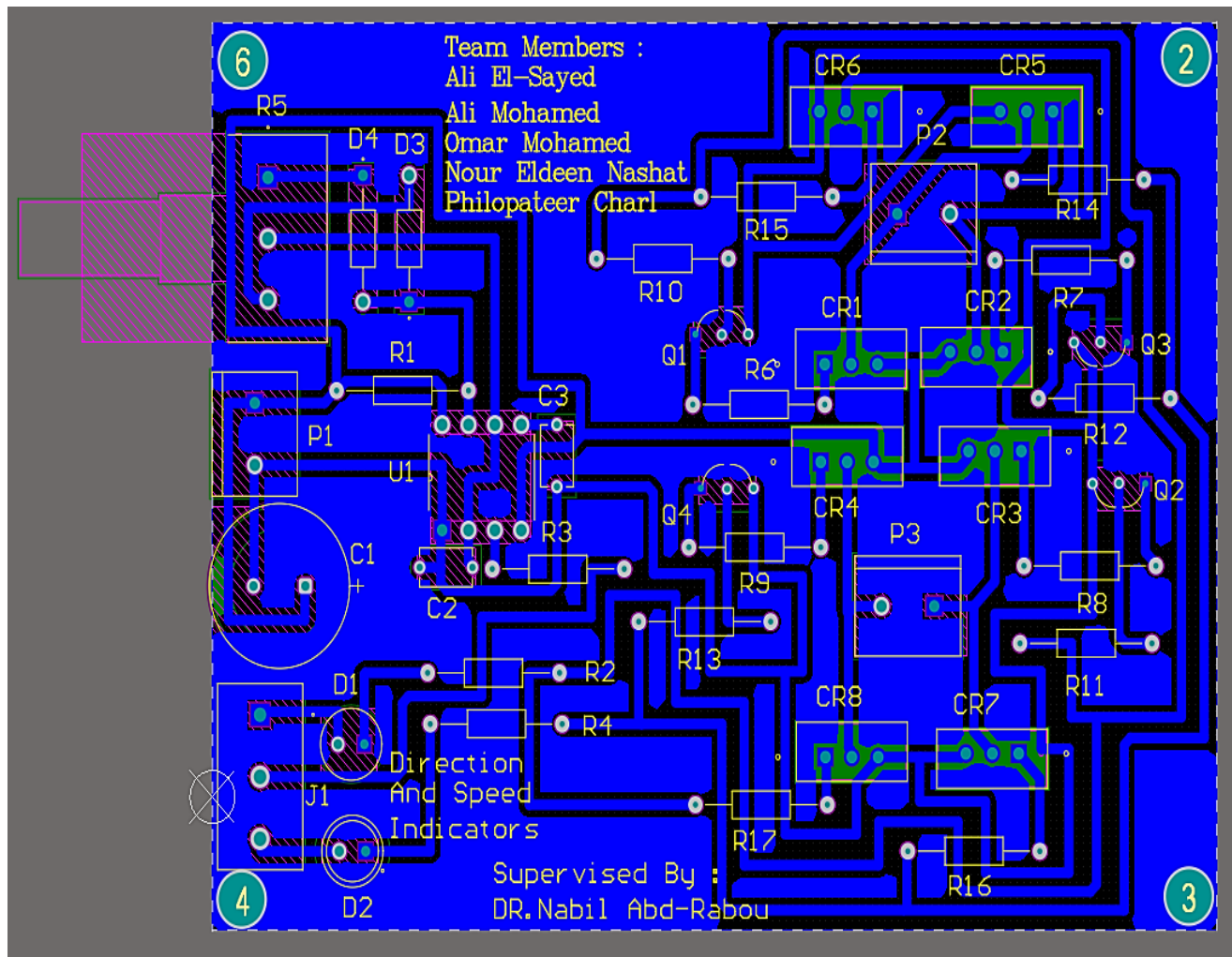


Figure 18 PCB Layout

After successful simulation, a printed circuit board (PCB) was designed using PCB layout software. Special care was taken to route the high-current motor paths separately from the control signals. The PCB was compact and suitable for mounting on a small robot or automation platform.

Now with no errors, it's ready for printing but we asked our Altium software for 3D visualization first and it also helped us with the PCB Final connection.

The Seventh Step

Ordering Our PCB

Order Number:	#8796	Order Number:	#8796
Date:	04/20/2025	Date:	04/20/2025
Total:	-	Total:	-
Status:	MANUFACTURING STARTED	Status:	COMPLETED
:		:	

Figure 21 PCB Order

Getting Our PCB Done

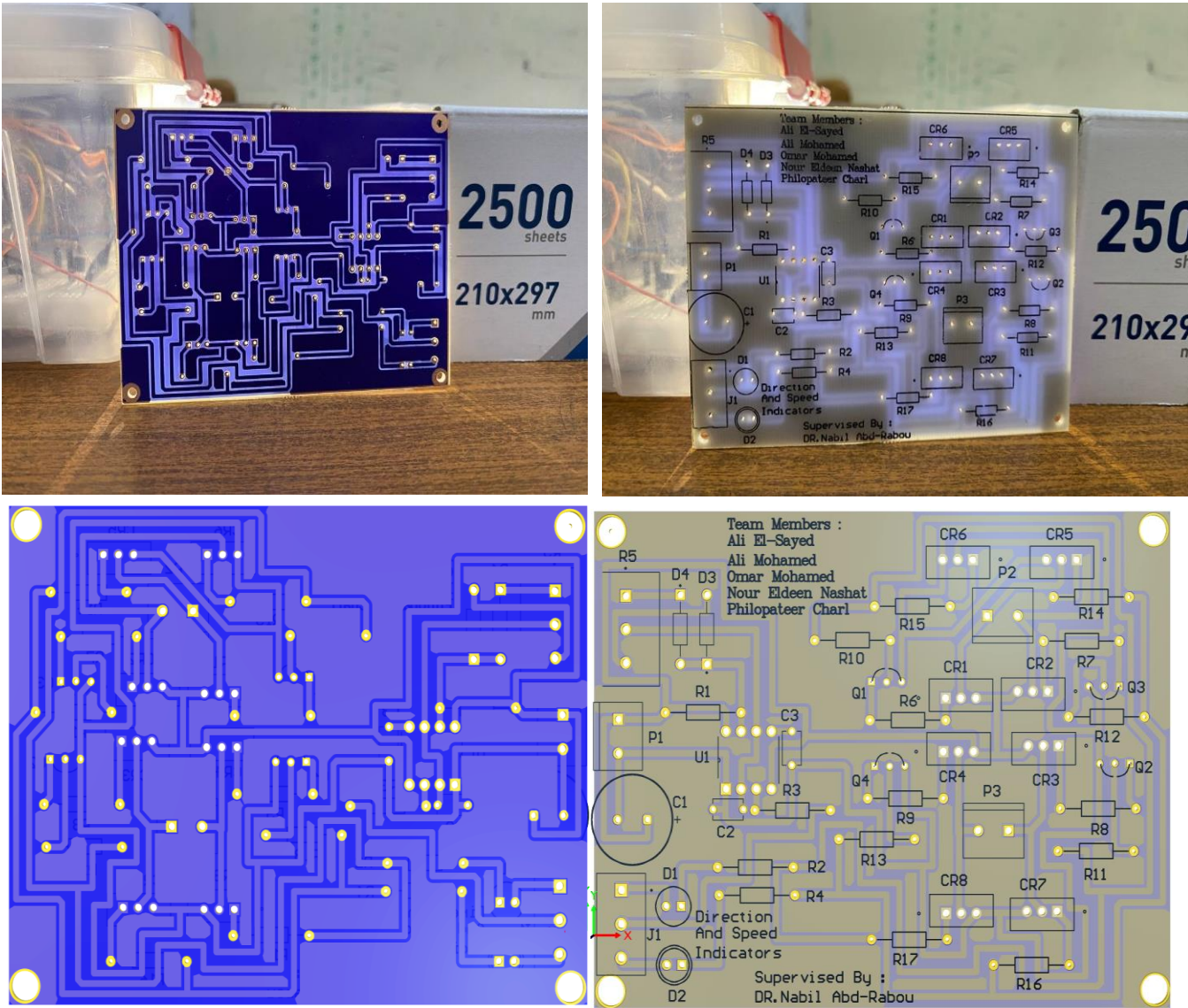


Figure 22 PCB Design VS Reality

The Eighth Step

Components Soldering

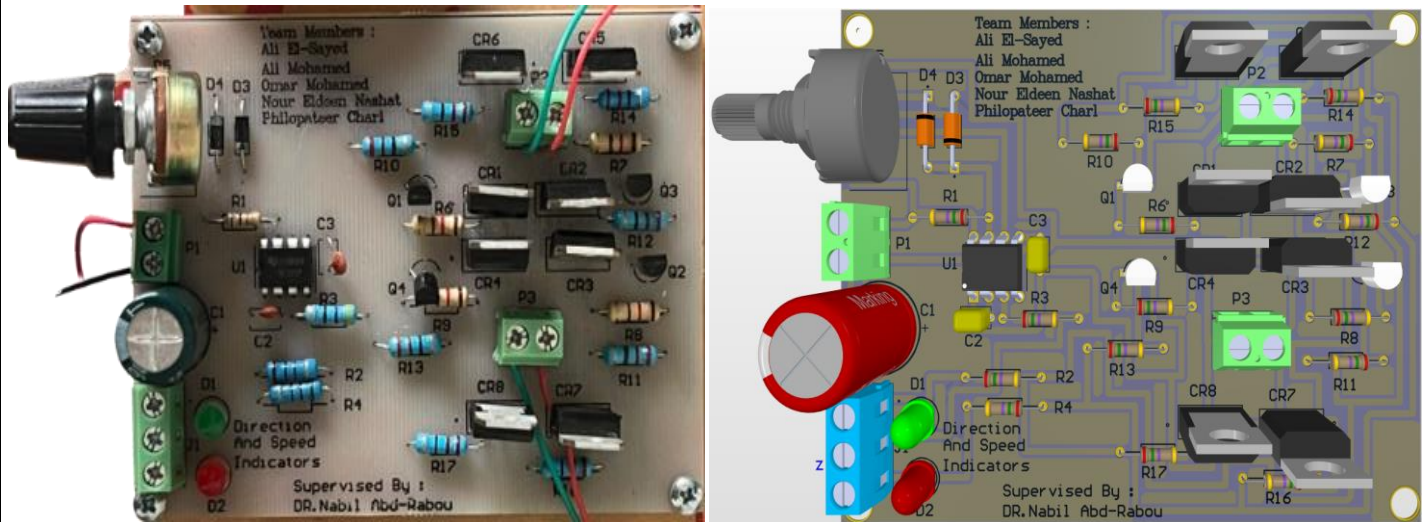


Figure 23 PCB After Soldering

The ninth Step

Finishing The project

We decided to finish our project in the form of a simple car robot

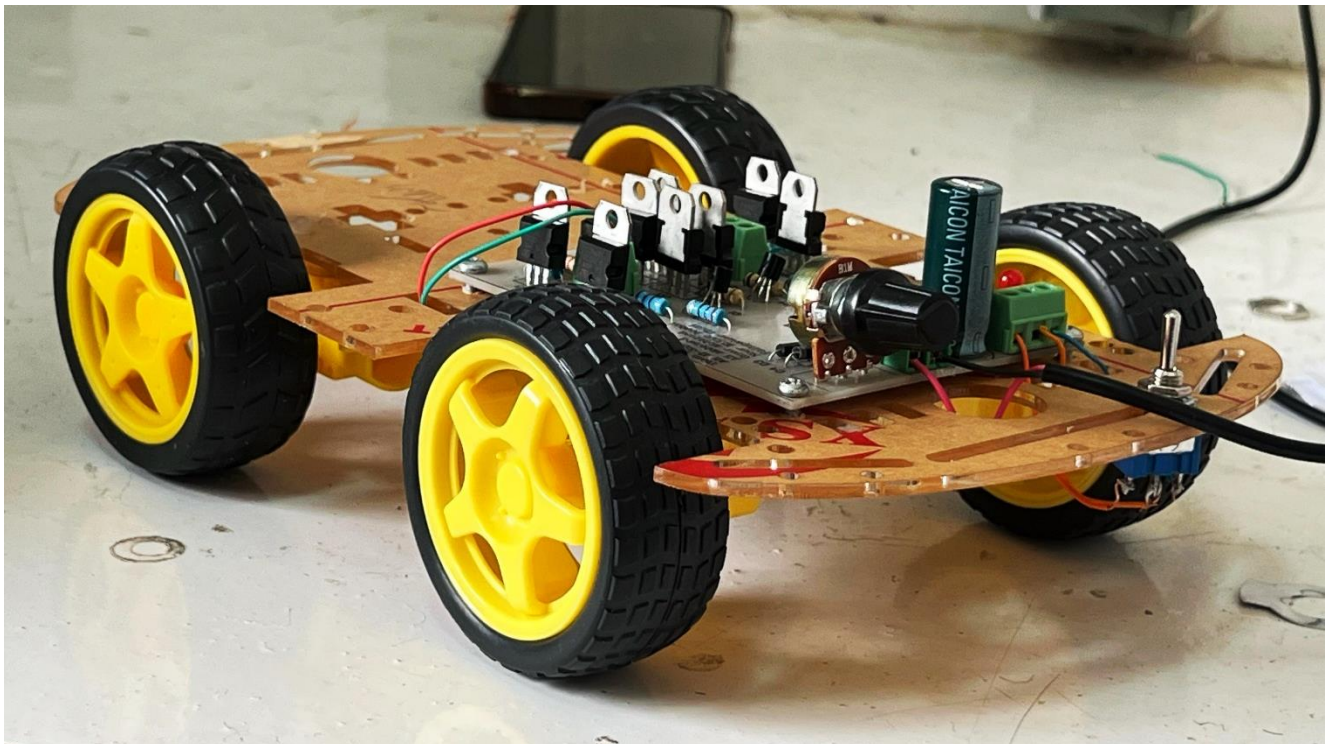


Figure 24 Simple Car Robot Application

This will keep our project open for any continuous Improvement like (Adding Wireless control system, adding cameras and memory to store the captured data, any other sensors for different Applications).

Advantages and Limitations

Advantages

- Cost-effective and simple implementation.
- No microcontroller or programming required.
- Independent control of two motors.
- Easily adjustable speed using a potentiometer.
- Useful for educational, hobby, and prototype projects.

Limitations

- **Simultaneous Control Only:** Cannot control the speed or direction of the motors independently.
- **No Feedback:** Operates open loop; actual motor speed may vary with load.
- **Shoot-through Risk:** Careful design and component matching are needed in the driver logic, to prevent simultaneous activation of high-side and low-side transistors on the same leg of an H-bridge.

Applications

- Simple differential drive robots (where turning is achieved by running motors in opposite directions or stopping one).
- Educational demonstrators for PWM and motor control principles.
- Basic automated systems require simple motion control (e.g., opening/closing a valve, simple conveyor).
- Animatronics or toys with basic movement.
- Electrical Drills & fans

Enhancements

Potential improvements to the circuit include:

- **Microcontroller Integration:** Use an Arduino/Raspberry Pi for automated control.
- Implementing speed feedback using encoders.
- Expanding the system for more than two motors.
- Adding Bluetooth or RF control for remote operation.
- Add a fuse in line with the motor power supply for protection.

Conclusion

This project successfully demonstrates a simple yet powerful method to control the speed and direction of two DC motors. Using a 555 Timer for PWM and an H-Bridge configuration for direction control provides a cost-effective and efficient solution for various robotics and automation applications. The system is modular, easy to build, and does not require programming, making it ideal for beginners and educational use.